

Modernizing Postgres Communication with Hackorum

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PostgreSQL Berlin Meetup · March 5th, 2026 · Zalando BHW

Who am I?

- **Kai Wagner** · Senior Engineering Manager at Percona
- I "bleed" open and I do work with and in Open Source for about 17y now.
- I worked on a variety of open source projects, including Ceph and the Linux Kernel at SUSE.
- I'm a licensed handball coach for over 20y as well as an agility and movement coach for kids

I joined the Postgres fun in 2022



postgresql github

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GitHub
https://github.com › postgres

PostgreSQL

Mirror of the official **PostgreSQL** GIT repository. Note that this is just a *mirror* - we don't work with pull requests on github.

Awesome Postgres awesome

A curated list of awesome PostgreSQL software, libraries ...

>

Postgres

Mirror of the official PostgreSQL GIT repository. Note that this is ...

>

[More results from github.com »](#)

Where are the issues and PRs?

postgres / postgres

<> Code

Agents

Actions

Security

Insights

Q Type to search

🔍

🔗

+

🔍

🔗

🔗

🔗

🔗

🔍

Watch 561

Fork 5.5k

Star 20.1k

🔍

Go to file

+

<> Code

🔍

Richard Guo

Fix unsafe RTE_GROUP removal in simplify_EXISTS_query

77c7a17 · 11 hours ago

63,392 Commits

📁 .github

Add CODE_OF_CONDUCT.md, CONTRIBUTING.md, and S...

2 years ago

📁 config

Restore AIX support.

2 days ago

📁 contrib

Fix variety of typos and grammar mistakes

yesterday

📁 doc

doc: Add link targets to CREATE/ALTER FOREIGN TABLE r...

yesterday

📁 src

Fix unsafe RTE_GROUP removal in simplify_EXISTS_query

11 hours ago

📄 .cirrus.star

ci: Simplify ci-os-only handling

6 months ago

📄 .cirrus.tasks.yml

meson: allow disabling building/installation of static libra...

2 days ago

📄 .cirrus.yml

ci: Per-repo configuration for manually trigger tasks

6 months ago

📄 .dir-locals.el

Make Emacs perl-mode indent more like perltidy.

8 years ago

📄 .editorconfig

Update .editorconfig and .gitattributes for postgresql.co...

3 months ago

📄 .git-blame-ignore-revs

Add a couple of recent commits to .git-blame-ignore-revs.

last month

📄 .gitattributes

Update .editorconfig and .gitattributes for postgresql.co...

3 months ago

📄 .gitignore

Update top-level .gitignore.

4 years ago

About

Mirror of the official PostgreSQL GIT repository. Note that this is just a *mirror* repository. - we don't work with pull requests on github. To contribute, please see https://wiki.postgresql.org/wiki/Submitting_a_Patch

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[www.postgresql.org/](https://wiki.postgresql.org/wiki/Submitting_a_Patch)

Readme

Code of conduct

Contributing

Security policy

Activity

Custom properties

20.1k stars

561 watching

5.5k forks

Report repository

Releases

679 tags

I'm confused! How do I contribute?

Want to edit, but don't see an edit button when logged in? [Click here](#).

Submitting a Patch

Contents [hide]

1 Initial patch design

1.1 Design your interface first

1.2 Save us the trouble of reformatting your code

2 Patch submission

2.1 Reasons your patch might be returned

2.2 Submission timing

3 Patch review and commit

4 Mutual Review Offset Obligations

5 Followup on submissions

5.1 How do you get someone to respond to you?

5.2 Submitting patch updates

Initial patch design

If you have a trivial patch that serves an obvious need, you may be able to write the patch and submit it directly to the [pgsql-hackers](#) mailing list without having its design reviewed first. But in general, a non-trivial change should be discussed (potentially before the code is even written) on the [pgsql-hackers](#) list before being submitted as a patch.

For general coding style guidelines, see the [Developer FAQ](#) and the [PostgreSQL Coding Conventions](#).

Design your interface first

Ask yourself these questions:

- Will the user interact with this new feature? if so, how?
- What is the syntax? Have ideas, and the ability to defend technical decisions you believe strongly in.
- What are the exact semantics/behaviors?
- Are there any backward compatibility issues?
- Get community buy-in at this level of detail before you start coding. But not necessarily consensus.
- Write an opening paragraph to your email to the -hackers list that answers these questions:
 - This is the kind of problem I'm trying to solve
 - This is what it is doing right now
 - This is what it will do.

Mostly, get someone from the community involved in your ideas as early as possible so that you can even get half-baked ideas vetted early, rather than creating something in a vacuum. Similarly, it's easier to make progress and keep patches focused if you concentrate on the smallest portion of the idea you can execute perfectly. Resist the temptation to build a giant patch all at once, as those are much less likely to be reviewed usefully and therefore committed. You should take a look at how your patch will eventually be [reviewed](#), so you can make sure that review is likely to succeed.

Save us the trouble of reformatting your code

Back to the roots! Mailing lists!



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- [Contributors](#)
- [Mailing Lists](#)
- [IRC](#)
- [Local User Groups](#)
- [Events](#)
- [International Sites](#)

PostgreSQL Mailing Lists

What are the PostgreSQL Mailing Lists?

The **PostgreSQL mailing lists** are one of the primary ways to interact with the PostgreSQL community including:

- Ongoing PostgreSQL development
- General support
- Product release & event announcements
- Coordinate community outreach

and many more.

Tips For Using the PostgreSQL Mailing Lists

To help ensure you have a productive experience on the PostgreSQL experience, we have a few tips

Tip #1: Read The Archives Policy

The PostgreSQL mailing lists are available in a public archive and are independently maintained by

There are....54 active ML's!?

View All Archives

User lists

List	Description
pgsql-admin	PostgreSQL administration (excludes pgAdmin application support, which can be found on the pgadmin-support list)
pgsql-advocacy	Coordinates people working on promoting PostgreSQL, organizing user groups, and dealing with PostgreSQL marketing and PR. Also a good place to ask for presentations and flyers if you need them.
pgsql-announce	Announcement list pertaining to PostgreSQL and various third party software.
pgsql-general	General discussion area for users. Apart from compile, acceptance test, and bug problems, most new users will probably only be interested in this mailing list (unless they want to contribute to development or documentation). All non-bug related questions regarding PostgreSQL's version of SQL, and all installation related questions that do not involve bugs or failed compiles, should be restricted to this area. Please note that many of the developers monitor this area.
pgsql-interfaces	Discussion of PostgreSQL interfaces, except JDBC and ODBC.
pgsql-jobs	Posting of PostgreSQL related jobs.
pgsql-novice	No question is too simple for this list.
pgsql-performance	Discussion of PostgreSQL's performance issues. Please see Guide to reporting problems and Slow Query Questions for some tips on how to write your performance question.
pgsql-sql	Discussion for users on SQL related matters.
pgsql-women	Discussion about how to increase the number of women in the PostgreSQL international community. All genders welcome.
Developer lists	
List	Description
buildfarm-members	Discussion list for buildfarm animal caretakers
pgsql-bugs	If you find a bug, please use the bug reporting form .
pgsql-committers	Notification of git commits are sent to this list. Do not post here!
pgsql-docs	Discussion regarding PostgreSQL documentation.
pgsql-gui-dev	Development of GUI tools for use with PostgreSQL
pgsql-hackers	The PostgreSQL developers team lives here. Discussion of current development issues, problems and bugs, and proposed new features. If your question cannot be answered by people in the other lists, and it is likely that only a developer will know the answer, you may re-post your question in this list. You must try elsewhere first!

This is a joke, right?

▼	pgsql	
■	pgsql-bugs	138
■	pgsql-general	736
■	pgsql-hackers	3,455

Can I request a 30h workday somehow?

Logical Replication of sequences

Started by Amit Kapila over 1 year ago 507 messages

 Merge Topic

Synchronizing slots from primary to standby

Started by Peter Eisentraut over 4 years ago 936 messages

 Merge Topic

The PostgreSQL mailing lists are the heartbeat of development.

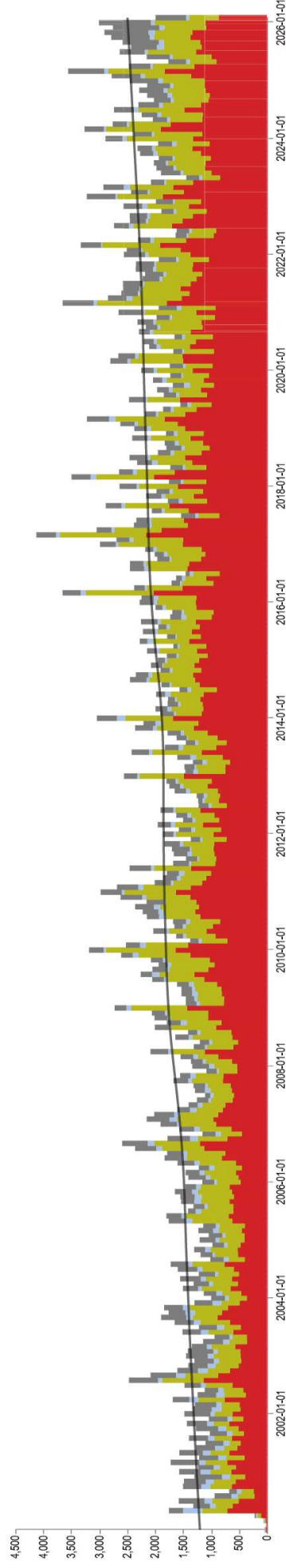
- `pgsql-hackers` - where PostgreSQL is actually built
- `pgsql-general`, `pgsql-docs`, `pgsql-bugs`
- **Almost 30 years of institutional knowledge** living in your inbox (yes since 1997)

"If it wasn't on -hackers, it didn't happen."

The Firehose Problem

▼ Messages

☒ Committers ☒ Contributors (non-committers) ☒ New participants ☒ Regular participants



- **pgsql-hackers:** 50–100+ messages *per day* (only on -hackers)
- Complex patches generate threads with **200–500+ messages**
- Multiple parallel discussions referencing each other

Pain Points

- **Context switching:** patch discussion $\leftrightarrow$ commitfest $\leftrightarrow$ CI
- **Lost threads:** "where was that discussion about logical replication locking?"
- **Barrier to entry:** new contributors don't know where to start
- **No collaboration layer:** teams can't annotate, tag, or track threads together
- **Search is terrible:** archive search is keyword-only, no boolean logic

Why We Can't Just Replace Email

- Decentralized by design - no corporate single point of failure
- Almost 30 years of archival history fully searchable
- Neutral ground - no vendor walled garden
- Established contributor culture and workflow expectations

*The goal: **evolve the interface, not the protocol***

Email is the source of truth. Hackorum is the lens.

[illegible]

- 

Main features

- Visualized complex conversations without losing your mind
- Commitfest Integration
- Contributor Profiles
- Patch Management
- Read Status
- Starring/Tagging/Mentions/Notifications
- Team Support
- Stats and Insights
- Import of read status and tags via CSV
- Mobile support

● LIVE DEMO of hackorum.dev

Did you have network issues? Click here!

Backup Slides

... otherwise we can go on :-)

What We Learned

Almost 30 Years of Email is Chaos

Modern standards don't apply to legacy archives. Replies arrive out of order, headers are missing (thanks for such great email clients in 2026), and some email clients marking all attachments as octet-stream...

The Rails 8 Surprise

After skipping several versions, the "out of the box" experience (Turbo, etc.) is a massive productivity boost for non-web devs.

Normalization is Too Slow

We tried for a "normalized" DB, but a feature-rich UI on a large dataset requires denormalization or caching for fast loads.

Agents are Force Multipliers

Coding agents have evolved rapidly and they are a speedup/help for modern web development.

I still hate CSS ;-)

The "POC" Gap

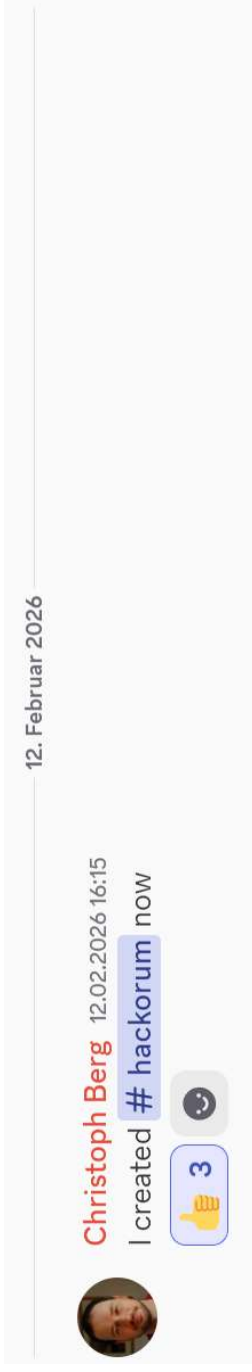
There is a massive difference between a "working demo" and a UI that actually handles the high-velocity activity of hackers.

Roadmap

- **Email reply from Hackorum** - close the loop (send replies back to the list)
- ~~**Lazy Loading**~~ ~~for the messages~~
- **Notification system** - get pinged on patches you care about (Like Slack/GitHub)
- **More lists** - postgresl-general, postgresl-docs
- **Git history parsing** - to collect merge info, to link to the actual commit in the thread

Get Involved

- Use it: hackorum.dev
- Code: github.com/hackorum-dev/hackorum
- Issues / Feature requests: [GitHub Issues](#)
- Feedback/Conversation: [PostgreSQL Hacking Discord](#)



Sorry for bumping old thread. And thanks to Hackorum, I now know this thread started exactly 10 years ago. After several patch attempts, it's still on my TODO list :)

Questions?

Find me after the talk or online:

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Slides: <https://imthekai.github.io/pgmeetup-berlin-march-2026/>



April 21-22 in Essen, Germany

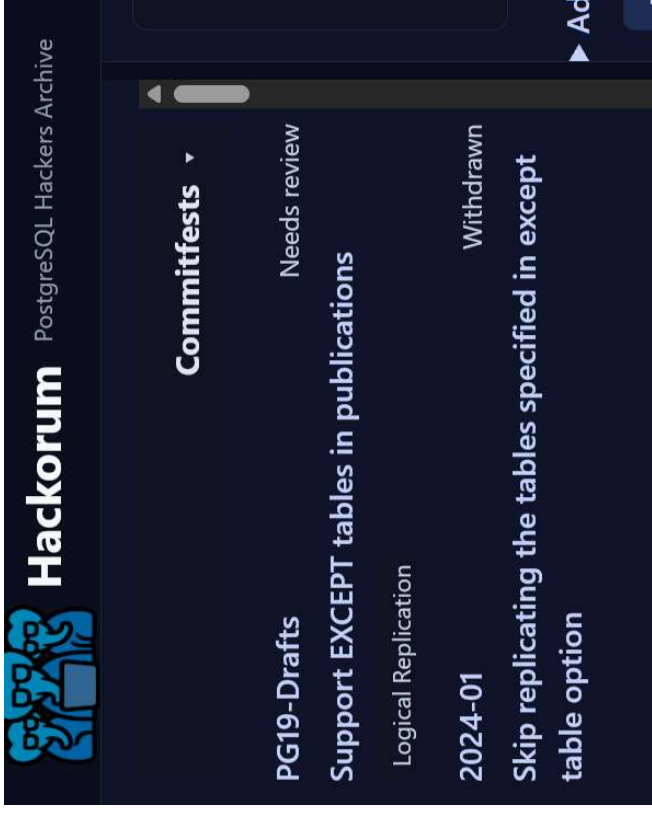
Backup Slides :-)

Feature Tour: Forum-Style Threads



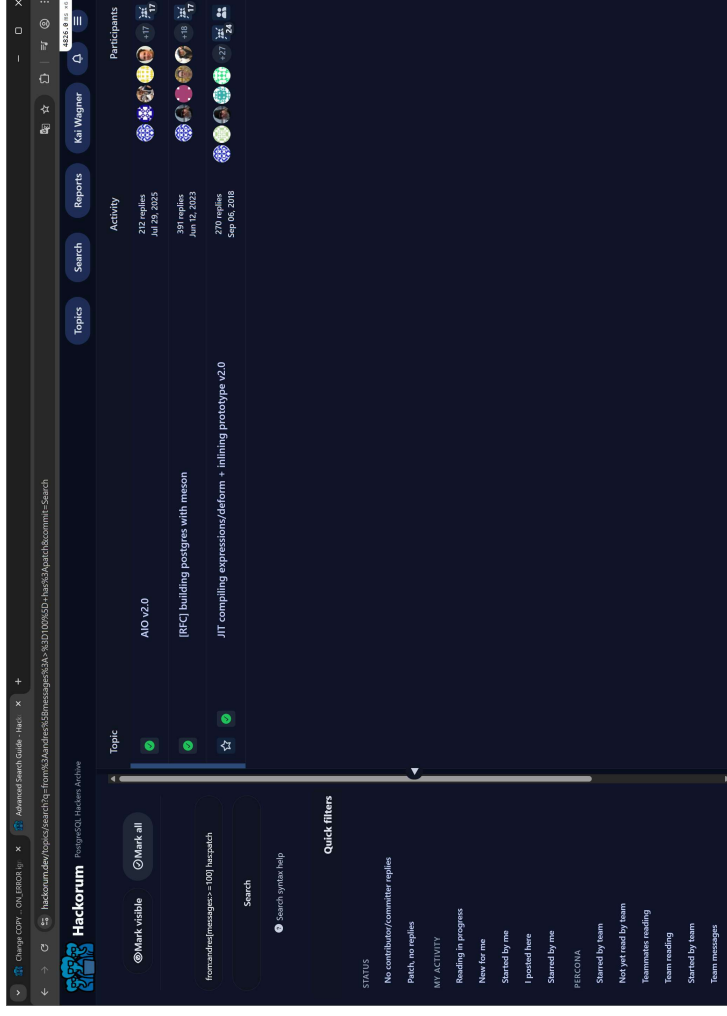
- Each mailing list thread = one **Topic**
- Messages displayed as nested threads (not flat chronological)
- **Per-message read tracking** - come back, pick up exactly where you left
- Thread merge for when discussions split/recombine

Feature Tour: Commitfest Integration



- **Live sync** with commitfest.postgresql.org (one a day)
- Patch status inline with the thread discussion:
 - Ready for Committer · Needs Review · Committed · Rejected
- CI scores + CI status visible per patch
- Reviewer and committer attribution
- Direct links from patch → discussion thread → commitfest

Feature Tour: Advanced Search



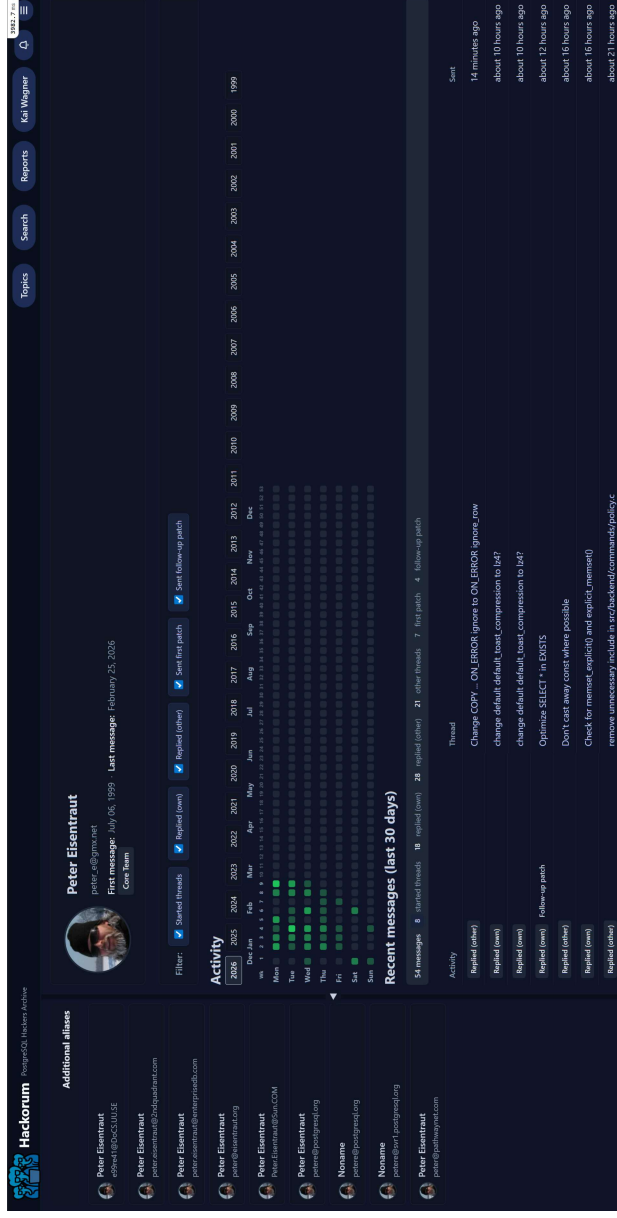
- **Full boolean search:** AND, OR, NOT
- **Field-specific queries:** `from:andres[message count >=100] has:patch` -> (More than 100 messages from Andres and has a patch attached)
- Filter by contributor tier, message count, participant count

Feature Tour: Teams & Collaboration



- **Teams** with private/public/open visibility
- Shared **notes** per topic - annotate discussions for your team
- @mention teammates and teams inside notes
- Note visibility scoped: author, mentioned users, mentioned teams
- Great for review assignments, patch tracking, onboarding

Feature Tour: Contributor Profiles



- **6-tier contributor hierarchy:**
 - Core Team - Committer - Major Contributor - Significant Contributor
 - Past Major or Significant Contributor
- Person identity resolution across **multiple email aliases**
- Per-alias send counts

Under The Hood: Tech Stack

Rails 8.0.2 · Ruby 3.4.4 · Hotwire/Turbo
Propshaft · ImportMap · Slim templates

PostgreSQL 18 (primary + 3 side DBs)
Solid **Queue** · Solid **Cache** · Solid Cable

IMAP IDLE → EmailIngestor → PG
Commitfest HTTP poller → PG

Kamal · Docker · Caddy · PgHero

No Redis. No Elasticsearch. No Sidekiq. All PostgreSQL.

Deep Dive: Person Identity Resolution

```
Email Alias A (kai@example.com)
Email Alias B (kai@percona.com)
Email Alias C (kaiwo@github.com)

└─ Contributor Membership → tier: :committer
```

```
# PersonIdPropagationJob: runs when alias → person assignment changes
class PersonIdPropagationJob < ApplicationJob
  def perform(person_id, old_person_id)
    ActiveRecord::Base.transaction do
      Message.where(sender_person_id: old_person_id)
        .update_all(sender_person_id: person_id)
      TopicParticipant.where(person_id: old_person_id)
        .update_all(person_id: person_id)
      # cleanup orphaned person record
      Person.find(old_person_id).destroy if old_person_id
    end
  end
end
```

Observability: PgHero + Rails Pulse



- **PgHero** - mounted at `/admin/pghero`
 - Slow query detection via `pg_stat_statements`
 - Index hit rates and bloat estimates
 - Live connection monitoring
- **Rails Pulse** - real-time performance monitoring dashboard

Deployment: Kamal + Docker

```
# deploy/config.yml (simplified)
service: hackorum
image: hackorum-dev/hackorum

servers:
  web:
    hosts: [production-host]
    cmd: bundle exec puma -C config/puma.rb

accessories:
  db:
    image: postgres:18
    volumes: [/data/postgres:/var/lib/postgresql/data]
  caddy:
    image: caddy:2
```

- **Kamal** handles zero-downtime deploys (blue/green containers)
- Postgres 18 in Docker with volume-mounted data
- Caddy as reverse proxy with automatic TLS
- Automated PG backups via sidecar container
- pg_stat_statements preloaded via Docker init script

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